The Gold Partition Conjecture Holds through Fourteen Elements

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Abstract

Peczarski verified the Gold Partition Conjecture for posets with at most 11 elements in 2006. We extend this exhaustive frontier to 14 elements. At order 14, the unique chain is set aside and every one of the remaining 1,338,193,159,770 isomorphism classes receives one of Peczarski's certificates. It follows, in particular, that the 1/3–2/3 Conjecture holds through order 14, one order beyond the previous complete mutual-rank-probability census. The calculation uses exact integer recurrences on the lattice of order ideals and was divided into 4,096 deterministic shards. The complete shard archive, source, and independent small-order checks accompany the paper.

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1 Introduction

Let P be a finite poset and write $\mathrm{e}(P)$ for its number of linear extensions. For $x, y \in P$, let $P + xy$ denote the poset obtained by adjoining $x < y$ and taking the transitive closure; if $y < x$ already, set $\mathrm{e}(P + xy) = 0$. Thus

$$p_P(x, y) = \frac{\mathrm{e}(P + xy)}{\mathrm{e}(P)}$$

is the probability that x precedes y in a uniformly chosen linear extension. For a non-chain P, its balance constant is

$$\mathrm{b}(P) = \max_{x \| y} \min\{p_P(x, y), p_P(y, x)\}.$$

The 1/3–2/3 Conjecture asserts that $\mathrm{b}(P) \geq 1/3$ for every finite non-chain. It was independently posed by Kislitsyn, Fredman, and Linial in connection with sorting under partial information [8, 7, 9]; see the survey of Chan and Pak [2].

The Gold Partition Conjecture is stronger. It asks for two comparisons chosen in advance such that, for every possible sequence of outcomes,

$$t_0 \geq t_1 + t_2,$$

where t_0 is the initial number of linear extensions and t_i is the number remaining after the first i comparisons. If the first comparison leaves a chain, Peczarski sets $t_2 = 1$. He introduced the

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conjecture, proved that it implies the 1/3–2/3 Conjecture, and verified it computationally through 11 elements [11, Conjecture 1, Proposition 1, and Theorem 3]. The name refers to the resulting sorting bound $C(P) \le \log_\varphi \mathrm{e}(P)$, where $\varphi = (1+\sqrt{5})/2$ [11, Propositions 2–3]. We are aware of no later exhaustive extension; Theorem 1 advances that frontier for the first time since 2006.

Theorem 1. *Every non-chain poset on at most* 14 *elements satisfies the Gold Partition Conjecture.*

Corollary 2. *Every non-chain poset on at most* 14 *elements has balance constant at least* 1/3.

Corollary 2 follows from Theorem 1 and Peczarski's Proposition 1. Only its order-14 case advances the previous computational frontier: De Loof, De Baets, and De Meyer computed all mutual rank probabilities and determined the worst balanced posets through order 13 [4]. The present calculation is a Gold Partition decision at order 14, not a census of all order-14 probabilities.

There has also been structural progress since Peczarski's original calculation. He proved the conjecture for every poset with a nontrivial automorphism, for a class containing all N-free posets, and for 6-thin posets [13, 12]. Dolores-Cuenca, Guzmán-Sáenz, and Kim proved that it is preserved by lexicographic sum (substitution) [5, Lemma 2.5]; their multiplicative argument applies unchanged to Peczarski's non-strict inequality.

2 Certificates

For an ordered incomparable pair (x, y), Peczarski calls z a *slave* if either $z > x$ and $z \parallel y$, or $z < y$ and $z \parallel x$. The verifier uses the following sufficient conditions from Peczarski's Definitions 1–2 and Lemmas 1–2 [11, pp. 91–92].

Lemma 3 (Peczarski)**.** *A non-chain P has the Gold Partition property if one of the following conditions holds.*

1. *There is an ordered incomparable pair (x, y) with at most one slave and*

$$2\mathrm{e}(P + xy) \ge \mathrm{e}(P).$$

2. *There are three distinct elements x, y, z such that*

$$\mathrm{e}(P + xy + yz) \le \max\{\mathrm{e}(P + yx), \mathrm{e}(P + zy)\} \le \frac{\mathrm{e}(P)}{2}.$$

The elements in condition (2) need not be pairwise incomparable. Peczarski's separate cyclic-triple screen is subsumed by the exhaustive search for condition (2). In each case the proof supplies two comparisons fixed before their outcomes are known, as required by the conjecture.

The verifier also accepts a half-balanced incomparable pair (x, y), meaning

$$2\mathrm{e}(P + xy) = 2\mathrm{e}(P + yx) = \mathrm{e}(P).$$

Indeed, use this pair as the first comparison. Then $t_1 = t_0/2$, while any fixed second comparison leaves at most t_1 extensions; hence $t_0 \ge t_1 + t_2$. This direct argument is also recorded by Peczarski [13].

There is a useful inexpensive screen for condition (1). If both orientations of an incomparable pair have at most one slave, one of $\mathrm{e}(P+xy)$ and $\mathrm{e}(P+yx)$ is at least $\mathrm{e}(P)/2$, so condition (1) holds without counting either quantity. The verifier applies this bilateral screen first. It then searches for a low-slave pair or a half-balanced pair, and finally for a triple satisfying condition (2). The division between the first two certificate counters depends on search order; their union does not.

3 Exact counting

Let $\mathcal{I}(P)$ be the lattice of order ideals of P, and let $F(I)$ be the number of linear extensions of the induced poset on I. Then

$$F(\varnothing) = 1, \qquad F(I) = \sum_{x \in \max(I)} F(I \setminus \{x\}). \tag{1}$$

Assigning zero to ideals that violate an adjoined comparison gives $\mathrm{e}(P+xy)$ and $\mathrm{e}(P+xy+yz)$.

After the first directly counted pair fails, a forward–backward pass supplies all pair counts. If $B(I)$ is the number of ways to complete a linear extension whose initial ideal is I, then

$$\mathrm{e}(P+xy) = \sum_{\substack{I \in \mathcal{I}(P) \\ x \in I,\ y \notin I \\ I \cup \{y\} \in \mathcal{I}(P)}} F(I) B(I \cup \{y\}). \tag{2}$$

Each product in (2) counts a subset of the linear extensions, their sum is at most $\mathrm{e}(P)$, and the comparisons use only small integer multiples. This ideal-lattice method for obtaining all mutual rank probabilities in one forward and one backward pass is due to De Loof, De Meyer, and De Baets [3]. Our contribution is not a new all-pairs algorithm; the implementation directs the established recurrences toward the certificates needed for the Gold Partition decision.

Candidate pairs are tried in a simple structural order. Candidate triples are ordered by decreasing right-hand bound in Lemma 3. Two cases require no new recurrence. If $z < x$, then $\mathrm{e}(P+xy+yz) = 0$. If $x < z$, the possible positions of y relative to $x < z$ partition the linear extensions and give

$$\mathrm{e}(P+xy+yz) = \mathrm{e}(P) - \mathrm{e}(P+yx) - \mathrm{e}(P+zy).$$

All extension counts are unsigned 64-bit integers. This is sufficient through order 14, since $\mathrm{e}(P) \leq 14!$; the formulas above use only small integer multiples of these counts.

The posets are supplied by Brinkmann and McKay's `genposetg`, which uses nauty for canonical labelling [1, 10]. The Gold Partition property is invariant under isomorphism, so one representative of each unlabeled class suffices. The production invocation, run once for each residue from 0 through 4095, was

```
gpc 14 o q m <residue> 4096.
```

Thus every unlabeled class was sent to the plugin, quiet output was requested, and the generator's deterministic modulus filter divided the stream into disjoint residues.

	$n = 12$	$n = 13$	$n = 14$
isomorphism classes	1,104,891,746	33,823,827,452	1,338,193,159,771
chain	1	1	1
pair certificate	1,066,006,204	32,418,324,910	1,272,077,147,789
balanced-triple certificate	38,885,541	1,405,502,541	66,116,011,981
open	0	0	0

Table 1: Gold Partition certificate aggregates.

4 Results and verification

Table 1 records the new orders. "Pair certificate" is the invariant union of the low-slave and half-balanced searches. For each order, the chain, pair-certificate, balanced-triple, and open rows partition the total.

The order-14 aggregate contains all 4,096 residues and agrees with the published number of unlabeled posets, first obtained by Heitzig and Reinhold and independently confirmed by Brinkmann and McKay [6, 1]. Exactly one class is a chain, the certificate counts partition the total, and the open count is zero. The known total checks enumeration completeness; the partition and shard checks are internal consistency tests. The zero open count is the computational conclusion. The released aggregator was rerun on the preserved 4,096-shard archive.

The same executable reproduces the known totals at orders 10 through 13 and the certificate counts of an earlier run. Independently, a Python program generates every unlabeled poset through order 7, enumerates its linear extensions, and applies the certificate definitions directly. A second implementation, using a plain subset recurrence, is compared with the released C classifier on every individual poset at orders 8 and 9, including non-topological labelings. The differential implementation shares neither extension-counting nor certificate code with the classifier. No independent per-poset check is made above order 9; at higher orders the evidence is the class total, the certificate partition, and the shard invariants. The companion full-pair program shares the ideal-lattice routines and reproduces the published counts of posets with linear-extension-majority cycles at orders 9 through 11 [4]. Tests of the shard protocol and aggregation rules are included with the source.

The order-14 calculation took approximately 1,594 core-hours and 18 hours of wall time on AMD EPYC 7B13 processors. In a controlled ablation on the production platform, removing only the bilateral early exit increased median user time by factors of 4.25, 4.81, 5.57, and 8.55 at orders 10 through 13, respectively. The first figure is a complete pass; the others aggregate the same three fixed modulus shards over alternating repetitions. The baseline binary is the production binary, and the total, pair-certificate union, balanced-triple count, and open count agreed in every comparison. This measures the screening step within the present implementation, not a speedup over the programs of Peczarski or De Loof et al.

5 Reproducibility

The manuscript, source, tests, and compact aggregate data are available at

`https://github.com/agupta/gold-partition-conjecture`.

The complete order-14 shard archive, its checksum manifest, build record, and the order-10–13 regression data are deposited at doi:10.5281/zenodo.21576030. The archive includes the two source files used to build the production classifier. The calculation used `genposetg` 1.1 from nauty 2.9.1 and a native GCC 13.3.0 build on AMD EPYC 7B13 processors. The repository build is portable by default. Its `make check` target runs the independent and differential small-order checks, aggregation tests, and an end-to-end sharding test.

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